

India boosts silicon photonics capability



Indigenously developed silicon photonics technology solutions at IIT Madras
(Credit: www.pib.gov.in)

India has launched two indigenous silicon photonics platforms to strengthen domestic capabilities in chip design, testing, and manufacturing for advanced photonic technologies. Developed at the MeitY-supported Centre of Excellence for CPPICS at Indian Institute of Technology Madras, the solutions include a Silicon Photonics Process Design Kit with over 50 verified components and a programmable photonic chip test engine. The technologies are designed to support startups, academia, the defence, and semiconductor industries in developing photonic integrated circuits. Research conducted in Chennai aligns India's silicon photonics capabilities with global standards while advancing the country's semiconductor self-reliance goals.

Autonomous robot for high-risk areas' inspections

Developed by the Hong Kong Productivity Council and partner institutions, an autonomous, AI-powered robot designed for nuclear facility inspections has been developed to improve safety and reduce human exposure in hazardous environments. Called Hazard-Bot, the system combines offline navigation, real-time mapping, AI perception, and a transformable wheeled mobility platform to inspect confined areas of nuclear plants without GPS or network connectivity. The robot demonstrated 96% accuracy in meter-reading tasks during trials at the Taishan Nuclear Power Plant in Guangdong. The technology supports continuous autonomous inspections and could also be deployed in water treatment plants, pumping stations, and other high-risk industrial facilities.

HazardBot, the autonomous transformable inspection robot developed by the Hong Kong Productivity Council and partners (Credit: www.hkpc.org)

